SAMPLE

Earth's Sky: Star Clusters

Pleiades, Hyades, and the Great Globulars

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Introduction

On a dark night, in a place the city lights have not yet reached, the sky offers a sight that rewards a second look: the patches of dim light that are not single stars but crowds. The Pleiades hang as a compact dipper of six or seven bright points wrapped in a faint glow; the Hyades spread their letter across the face of the bull; the Beehive glows as a soft spot the naked eye resolves into a dozen glints with the least optical help; and far beyond all of them, in a halo we did not know we had, the great globulars burn as balls of ten thousand stars at once, the most distant steady objects the naked eye ever holds. These are the star clusters, and this volume is their full account: the crowds of stars born together, aging together, and traveling together, the open clusters loose and young in the galaxy's disk, the globulars dense and ancient in its halo, and the whole ladder of learning the crowds have given astronomy, from the stellar distance scale to the age of the universe.

The reader should also know the crowds' own variety before the chapters begin: the youngest open crowds still glow in their birth nebulosity, the young Pleiades ride free of theirs, the middle-aged Hyades and Beehive spread wide across the winter sky, the oldest open crowds hang as survivors in the plane, and the globulars sit above and below the whole map in their halo orbits, so a single year of watching meets the family at every age, the book's own organizing sight, and the chapters follow the same arc from the youngest to the oldest, the catalog's standing structure.

The reader should know at the outset why clusters are the astronomer's favorite tool as much as the watcher's favorite sight. A cluster is a controlled experiment the sky ran on its own: its stars formed from one cloud at one time, so they share an age, a chemical makeup, and a distance from Earth, which means the differences among them are real physical differences, not the effects of different origins. The astronomer who studies a cluster is holding a natural cross-section of the stellar life cycle, every stage from the newly lit to the nearly spent sitting side by side at a known remove, and the chapters of this book explain how that single fact has built a surprising share of modern astronomy, the distance ladder, the stellar ages, and the galaxy's own shape included.

The conventions' note belongs in the preview's close: this volume treats the crowds as families, not as individual stars, the members' own biographies living in the stellar siblings the shelves hold, and the reader who wants the bright sisters' names and the giant vaults' variable clocks one by one will find the cross-links placed through the chapters, the series' standing division of labor, the book's own scope held plainly.

The preview paragraph, in the catalog's standing style: the open clusters are the disk's young families, hundreds of stars wide, dissolving into the general star field within a few hundred million years, the Pleiades and the Hyades the flagship examples; the globulars are the halo's ancient vaults, hundreds of thousands of stars packed into spheres nearly as old as the universe itself, the Hercules cluster and the southern showpieces the flagship examples; the stars within each crowd are siblings by birth and composition, so a cluster's membership can be checked by measurement, and the whole crowd can be dated by its brightest survivors; the clusters serve as standard candles, distance benchmarks, and clocks, the roles the research chapters describe; and the best of them are among the easiest things in the sky to find and among the finest a telescope of any size can show. The chapters unpack each clause, with the finding charts and the honest warnings every volume in the catalog carries.

Chapter 1: The Discovery of Star Clusters

The discovery story of the clusters runs from the oldest records to the newest catalogs, and it divides cleanly at the telescope, because the crowds of stars are the sky's oldest listed objects and its longest-continuing discoveries at once.

The oldest record's own texture deserves the first paragraph: the Pleiades' dawn rising, the group's appearance before the sun in the late spring, set the ancient farming calendars, the year's plowing and the sailing seasons keyed to the crowd's own return, and the Pacific navigators read the group's dawn risings for their new year, the same clock in both hemispheres, the oldest record being a working calendar's, not a stargazer's, the discovery story's own first lesson that the clusters were tools before they were objects.

The naked-eye clusters come first, in the oldest records we have. The Pleiades appear in the earliest written sky lists and in the oral calendars of cultures on every continent, a compact group too prominent to overlook and too regular to ignore, the year's farming clock in the ancient Mediterranean, the Maori new year's marker in the Pacific, and the Subaru of the eastern navigators whose name now rides on a car badge, a cluster famous in every hemisphere that watched it. The Hyades ride beside it with the bull's face, the V of the nearest cluster we know;

the Beehive hung in the crab as a misty patch the old watchers logged as a single cloudy star; and the star-poor southern skies kept the two great globulars, the largest of them all hanging beside the southern cross, a naked-eye fuzz the travelers' logbooks noted without understanding. The pre-telescopic record is a register of puzzles: objects catalogued as clouds, counted as single stars, or praised as omens, all of them clusters awaiting resolution.

The in-between era's record deserves its line: the classical catalogs logged the cloudy objects as single fixed entries, the medieval star lists carried them unchanged, and the early telescopic age's own catalog, the Sicilian astronomer's list of nebulous objects from the generation before the famous comet hunters, counted the Beehive and the Pleiades' nebulosity among its entries, the record's honest note that the resolution was coming and the counting was ready for it.

The telescope's arrival, in the age of the first lenses, began the resolution. The earliest telescopic observers turned the new instruments on the misty objects and found stars: the Beehive split into dozens of points, the Pleiades gained their fainter sisters, and the naked-eye clouds began their long conversion into catalogs of stellar membership. The great sweeps of the following era, the comet hunters who needed to know which fuzzy patches were their quarry and which were standing sky furniture, produced the famous list of fixed objects to ignore, a list that begins with a comet impostor in Taurus and runs through the clusters and nebulae the modern sky still carries under those same numbers, and the century's telescope-builders, sweeping with mirrors of unprecedented size, resolved the deeper crowds and found the globulars by the dozens, the sky's ball-shaped patch inventory rising from a handful to the modern count.

The southern giant's own discovery deserves its line in the story: the largest globular was logged by the southern travelers as a star of the southern sky, named as a single point in the early catalogs, and its true nature as the sky's greatest star crowd came only with the telescopic age's resolves, the naming history the trivia chapter enjoys, a whole galaxy's worth of suns catalogued first as one star, the discovery story's own best compression of the whole subject's arc.

The century that followed turned the clusters from furniture into tools. The astronomers who sorted stars by their true brightness found that a cluster's stars, plotted on the famous diagram of color against light, trace the stellar lifespan in a single sweep, the turnoff point reading the crowd's age; the variable-star hunters found the globulars full of pulsing standards whose periods reveal their distances, and the young astronomer who mapped those distances found the globulars arrayed around a center far from our own position, the discovery that relocated the sun within the galaxy, a discovery the discovery chapter of this book describes as the clusters' greatest gift. The modern era added the space telescopes' resolves, the membership surveys by the million, and the working realization that clusters are not eternal: the young crowds dissolve, the old ones evaporate, and the catalogued crowds are survivors of a much larger birth rate, a modern understanding the later chapters build on.

The chapter's summary, in the series' plain style: the clusters are the sky's oldest listed objects, resolved by the telescope, organized into catalogs by the comet hunters, turned into tools by the diagram-plotters and the variable-star timers, and still being counted today, a discovery story in which the reader's own binoculars stand on the same ladder as the great surveys.

A paragraph on the first cluster anyone ever met, the little group that tests the eye. The subject of the file begins with the sky's most famous family, the tight knot of young hot stars near the bull's shoulder, known to every sky culture as sisters, doves, or a handful of seeds, and famous for a quality no other cluster shares: it is a self-administered eye exam, most people seeing six of its stars, sharp-eyed viewers counting seven or more, the little group having served as a vision test long before it served as astronomy; the family is also the standard first cluster of every telescope, resolving into a dozen bright members plus a haze of fainter ones in any glass. The opening note worth recording: the sisters' double fame, as naked-eye riddle and as telescopic showcase, makes them the file's natural front door, and the ancients' counting games around them count as cluster astronomy in its folk form, a census taken eye by eye, millennia before anyone knew the counted stars were true siblings, born together, moving together, the sky's oldest membership list kept by campfire and contest.

A second opening card, on the night the smudges opened, the telescope's first family reunion. When the early telescopes turned on the sky's smudges, several of them dissolved into swarms: the famous cloud of the cancer, the beehive, resolved from a fog to a crowd; the sisters gaining a dozen new members in the first look; the dense little balls of the northern sky showing as grainy spheres, on the edge of resolution, the glass proving that the sky's dim patches were largely crowds, the first generation of telescopes earning their keep by splitting families into individuals. The closing note for the opening chapter: the telescope's first great discovery was membership, the smudge becoming a census, and the file's whole history inherits that pattern, every new instrument being judged,

in part, by how far it can take the resolution of the crowds, the sisters' first glass and the beehive's first look standing at the head of a lineage that runs to the modern surveys counting millions of members, the sky's smudges having turned out to be, again and again, the sky's neighborhoods.